1 Define CFG and design ~~CT~~ CFG for the following

Ans Context Free Grammar (CFG) is a formal grammar that uses a set of production rules to generate strings in a formal language.

~~a) $L = \{0^{2n} 1^m \mid n>=0, m>=0\}$~~

Ans $G = (V, T, S, P)$

where V - Finite set of Variable

T - Finite set of Terminal

S - Start symbol

P - Set of production rules

a) $L = \{0^{2n} 1^m \mid n>=0, m>=0\}$

Ans $L = \{\varepsilon, 1, 11 \ldots, 00, 0000, \ldots, 001, 00001 \ldots, 0011, 00111 \ldots\}$

$S \to \varepsilon \mid 00 \mid 1$

$S \to 00S \mid S1$

Derivation :- 00001

$S \Rightarrow 00S$

$\Rightarrow 0000S$

$\Rightarrow 00001$

$G = (V, T, S, P)$

$V = \{S\}$

$T = \{0, 1\}$

$S \to \varepsilon \mid 00 \mid 1 \;/\; 00S \mid S1$

$P = S \to \varepsilon \mid 00 \mid 1$

$S \to 00S \mid S1$

b) $L = \{0^i \cdot 1^j \, 2^k \mid i=j \text{ or } j=k\}$

Ans * $i=j$

$L = \{\varepsilon, 2, 012, 00112 \ldots\}$

$S \to \varepsilon \mid 2 \mid 01$

~~$S \to 0S1S \mid$~~

$S \to 0S1S \mid S2$

Derivation :- 0011222

$S \Rightarrow 0S1S$

$\Rightarrow 0011S$

$\Rightarrow 0011S2$

$\Rightarrow 001 \cancel{1} S22$

$\Rightarrow 0011222$

$V = \{S\}$

$T = \{0, 1, 2\}$

$S \to \varepsilon \mid 2 \mid 01 \mid 0S1S \mid S2$

$P = S \to 0S1S \mid S2$

$S \to \varepsilon \mid 2 \mid 01$

* $j=k$

$L = \{\varepsilon, 0, 012, 0012 \ldots\}$

$A \to \varepsilon \mid 0 \mid 12$

$A \to 0A \mid 1A2$

Derivation :- 0012

$A \Rightarrow 0S$

$\Rightarrow 00S$

$\Rightarrow 0012$

$V = \{A\}$

$T = \{0, 1, 2\}$

$A \to \varepsilon \mid 0 \mid 12 \mid 0S \mid 1S2$

$P = A \to \varepsilon \mid 0 \mid 12$

$A \to 0S \mid 1S2$

$S \to 2 \mid 01 \mid \varepsilon$

$A \to 0 \mid 12 \mid \varepsilon$

$B \to 0S1S \mid S2 \mid 0A \mid 1A2$

c) $L = \{a^n b^{2n} \,.\, \Sigma = \{a, b\} \; n \geq 1\}$

Ans $L = \{abb, aabbbb \ldots\}$

$S \to abb \mid aSbb$

Derivation :- aabbbb

$S \Rightarrow aSbb$

$\Rightarrow aabbbb$

d) $L = \{0^m 1^m 2^n \mid m \geq 1, n \geq 0\}$

A) $L = \{01, 0011 \ldots, 012, 0122, 001122 \ldots\}$

$S \to 01A$
$S \to 0S1A$
$A \to \varepsilon \mid 2A$

~~$S \to 0S1 \mid A$~~
~~$A \to 2A \mid \varepsilon$~~

Derivation : 0122

$S \Rightarrow 01A$
$\Rightarrow 012A$
$\Rightarrow 0122$

e) $L = \{ww^R \mid w \in (a,b)^*\}$

A) $L = \{aa, aaaa \ldots, bb, bbbb \ldots, abba, abaaba, \ldots\}$

$S \to aSa \mid bSb \mid \varepsilon$

w = aba

Derivation : abaaba

$S \Rightarrow aSa$
$\Rightarrow abSba$
$\Rightarrow abaSaba$
$\Rightarrow abaaba$

f) $L = \{a^n b^m c^k \mid n + 2m = k \;.\; m \geq 0, n \geq 0\}$

A) ~~$S \to aSc \mid bSCC \mid \varepsilon$~~

$L = \{\varepsilon, ac, aacc, bcc, abccc \ldots\}$

$S \to aSc \mid bSCC \mid \varepsilon$

Derivation :- abbccccc

$S \Rightarrow aSc$
$\Rightarrow abSccc$
$\Rightarrow abbSccccc$
$\Rightarrow abbccccc$

g) $L = \{w : |w| \bmod 3 = 0\}$ $\Sigma = \{a\}$

A) $L = \{\varepsilon, aaa, aaaaaa \ldots\}$

$S \to aaaS \mid \varepsilon$

Derivation aaaaaa

$S \Rightarrow aaaS$
$\Rightarrow aaaaaaS$
$\Rightarrow aaaaaa$

2 Define ambiguity. Find LMD, RMD and parse tree for the following. Also show if the grammar is ambiguous or not

A Ambiguity in context-free grammars occurs when at least one string can be generated by the grammar in more than one way - that is, it has two or more different leftmost derivations (LMD), rightmost derivation, or parse trees

a) $S \to AbB$ - (1)
$A \to aA \mid \varepsilon$ - (2)
$B \to aB \mid bB \mid \varepsilon$ (3)
$D \to a \mid \varepsilon$ - (4)

string : "aaabab"

Ans (1) ~~Taking 'A' as priority~~

LMD

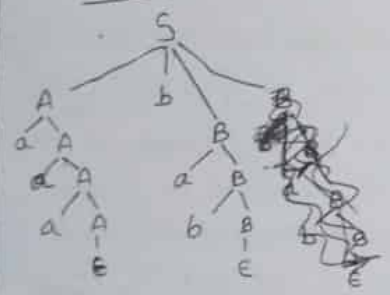
S => AbB - (1)
=> aAbB - (2)
=> aaAbB - (2)
=> aaaAbB - (2)
=> aaabB - (2)
=> aaabaB - (3)
=> aaababB - (3)
=> aaabab - (3)

RMD

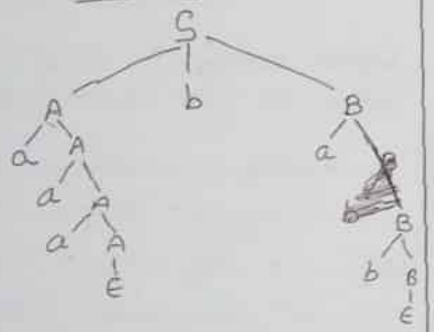
S => AbB
=> AbaB - (3)
=> AbbaB - (3)
=> AbabB - (3)
=> Abab - (3)
=> aAbab - (2)
=> aaAbab - (2)
=> aaaAbab - (2)
=> aaabab

-> Parse tree

LMD



RMD



~~LMD (Taking B as priority)~~

~~S => AbB~~

Since their is no difference b/w Parse tree
∴ It is not Abiques

b) E -> E+E | E-E | E*E | E/E | a | b
String 'a+b*a+b'

Ans

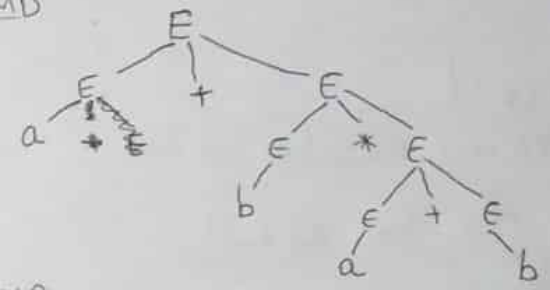
LMD

S => E+E
=> a+E
=> a+E*E
=> a+b*E
=> a+b*E+E
=> a+b*a+E
=> a+b*a+b

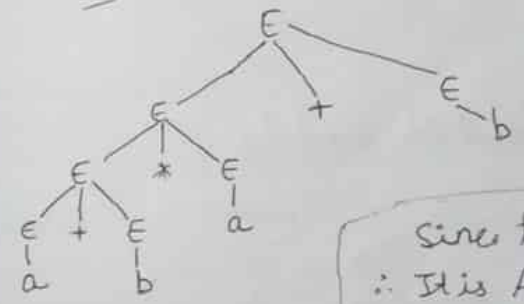
RMD

S => E+E
=> E+b
=> E*E+b
=> E*a+b
=> E+E*a+b
=> E+b*a+b
=> a+b*a+b

LMD



RMD



Since their are 2 different Parse tree
∴ It is Abiques

3 Define CNF, convert the following CFG to CNF

A ~~CNF is a formal grammer that uses a set of production rules to generate strings in a formal language.~~

a) $S \to aACa$
$A \to B|a$
$B \to c|\varepsilon$
$C \to cC|\varepsilon$

Chomsky Normal form (CNF) is a standardized way to write context-free grammars where every production is either
$A \to BC$
$A \to a$
$S \to \varepsilon$

Ans (i) Eliminate unwanted symbols

* Generative	Non-Generative
$C \to \varepsilon$	
$A \to a$	
$B \to c$	
$S \to aa\varepsilon a$	

∴ Their is no non-Generative symbols

* Reachable	Non-Reachable
S	
A	
C	
B	

∴ Their is no Non-Reachable symbols

(ii) Eliminate 'ε'
$C \to \varepsilon$
$S \to \cancel{aACa}\ aAa$
$A \to B|a$
$B \to c$
$C \to cC$

(iii) Unit production
$A \to B$

$B \to c$
$A \to c|a$
$C \to c$
~~$S \to accca|acca|aaca|aca|aaa$~~
$S \to aca|aaa$

(iv) ~~$A \to BC$ or $A \to a$~~ $A \to BC$ or $A \to a$

~~$B \to c$~~
~~$A \to c|a$~~
~~$C \to c$~~
~~$C' \to c$~~
~~$S \to ZA'|$~~
~~$C^2 \to$~~
~~$S \to ZA'|YA'|XA'|A'W|A'W|A'A^2$~~

$B \to c$
$A \to c|a$
$C \to c$
$S \to A'D|A'A^2$

$S \to A'$
$A' \to a$
$A^2 \to A'A'$
$D \to CA'$

$A' \to a$
$A' \to A'A'$
$A^2 \to A'A'$
$D \to CA'|BA'$

~~$A' \to a$~~
~~$A^2 \to A'A'$~~
~~$C^2 \to C'C'$~~
~~$C^3 \to C^2C'$~~
~~$Z \to A'C^3$~~
~~$Z' \to ZA'$~~
~~$Y \to AC^2$~~
~~$Y' \to YA'$~~
~~$X \to A^2C^2$~~
~~$W \to ca$~~

b) $S \to ASB \mid \varepsilon$
$A \to aAS \mid a$
$B \to SbS \mid A \mid bb$

A (i) Eliminate unwanted symbols

*

Generative	Non-Generative
$S \to \varepsilon$ $A \to a$ $B \to bb$	ϕ

$\therefore$ There no Non-Generative symbols

$S \to ASB \mid \varepsilon$
$A \to aAS \mid a$
$B \to SbS \mid A \mid bb$

*

Reachable	Non-reachable
S A B	

$\therefore$ There is no Non-reachable symbols

$S \to ASB \mid \varepsilon$
$A \to aAS \mid a$
$B \to SbS \mid A \mid bb$

(ii) Eliminate ε production

$S \to \varepsilon$

$A \to aA \mid a$
$B \to bS \mid Sb \mid A \mid bb$
$S \to AB$

(iii) Eliminate unit production

$B \to A$

$A \to aA \mid a$
$B \to bS \mid Sb \mid aA \mid a \mid bb$ $\quad S \to AB$

(iv) $A \to BC$ or $A \to a$

$S \to AB$
$A \to aA'A \mid a$
$B \to B'S \mid SB' \mid A'A \mid a \mid B'B'$

$A' \to a$
$B' \to b$

c) $S \to ABC \mid BaB$
$A \to aA \mid Bac \mid aaa$
$B \to bBb \mid a \mid D$
$C \to CA \mid AC$
$D \to \varepsilon$

A (i) Eliminate unwanted symbols

*

Generative	Non-Generative
$D \to \varepsilon$ $B \to a \mid \varepsilon$ $A \to aaa$ $S \to a$	$C \to CA \mid AC$

Eliminat Non-Generative

$S \to BaB$
$A \to aA \mid aaa$
$B \to bBb \mid a \mid D$
$D \to \varepsilon$

(ii) *

Reachable	Non-reachable
S B D	$A \to$

b) Eliminate A

$S \to BaB$

$B \to bBb \mid a \mid D$

$D \to \varepsilon$

(ii) Eliminate ε

$D \to \varepsilon$

$S \to BaB$

$B \to bBb \mid a$

(iii) Eliminate unit production

∴ There is no unit production

$S \to BaB$

$B \to bBb \mid a$

(iv) $A \to BC$ or $A \to a$

$B \to B^2B' \mid a$

$S \to B^3B$

$B' \to b$

$B^2 \to B'B$

$B^3 \to BA'$

$A' \to a$

4 Define PDA. Construct PDA for the following

Ans Pushdown Automaton (PDA)

- A PDA is defined by states, transitions, an input alphabet, a stack alphabet, a transition function, a start state, a set of accepting states, and a stack with push/pop operations
- The stack allows the automaton to store and retrieve data, enabling it to handle languages beyond the power of DFAs/NFAs

PDA :- can be defined as 7 components

$P' = (Q, \Sigma, \vdash, \delta, q_0, Z_0, F)$

$Q \to$ Finite set of states

$\Sigma \to$ i/p symbols

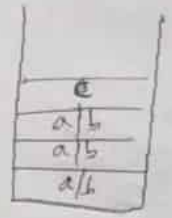
$\vdash \to$ stack symbol

$q_0 \to$ start state

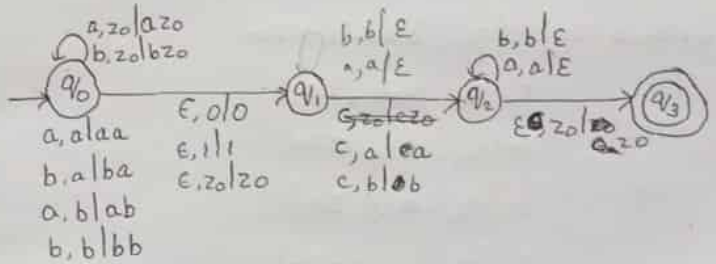
$Z_0 \to$ start symbol in stack

$F \to$ set of Final states

$\delta \to \hat{\delta}(q, a, x) = (P, Y)$



a) $L = \{wcw^R \mid w \in (a,b)^*,\ w^R$ is reverse of $w\}$ show moves made by PDA for the string "aabcbaa" and "abacbba"



(i) aabcbaa

C reaches

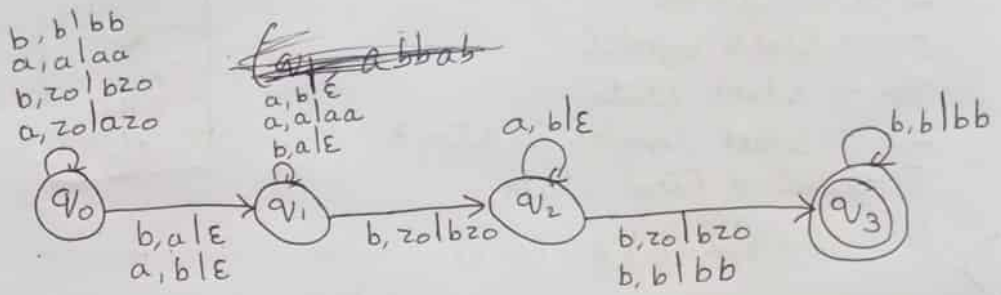
∴ It is accepted

(ii) abacbba

∴ It is not accepted

b $L = \{ n_a(w) < n_b(w),\ w \in (a+b)^* \}$
Show ID for the string abbab

Ans



ID:-

$(q_0, abbab, z_0) \vdash (q_0, bbab, az_0)$
$\vdash (q_1, bab, z_0)$
$\vdash (q_2, ab, bz_0)$
$\vdash (q_2, b, z_0)$
$\vdash (q_3, \varepsilon, bz_0)$